

EX:NO:25

PROGRAM:

```
import math
```

```
def closest_pair(points):
```

```
    min_dist = float('inf')
```

```
    pair = None
```

```
    for i in range(len(points)):
```

```
        for j in range(i+1, len(points)):
```

```
            dist = math.dist(points[i], points[j])
```

```
            if dist < min_dist:
```

```
                min_dist = dist
```

```
                pair = (points[i], points[j])
```

```
    return pair, min_dist
```

```
# Test
```

```
points = [(1,2),(4,5),(7,8),(3,1)]
```

```
print(closest_pair(points))
```

```
# ((1,2),(3,1)), 1.4142...
```

Output

```
((1, 2), (3, 1)), 2.23606797749979)
```

```
=== Code Execution Successful ===
```